Quizrr Chapter-Wise Test for JEE Advanced - 2025

By: C I P H Ξ R

Q.1

If $\left(\sin^{-1} x\right)^2 + \left(\cos^{-1} x\right)^2 = \frac{5\pi^2}{8}$, then find the value of x.

(A) $-1/\sqrt{2}$

(B) $\pm 1/2\sqrt{2}$

(C) $\pm 1/2$

(D) $-1/3$

Q.2

$\tan^{-1}(1/3) + \tan^{-1}(1/7) + \tan^{-1}(1/13) + \ldots + \tan^{-1}\left(\frac{1}{n^2+n+1}\right) + \ldots \infty$ is equal to

(A) 0

(B) $\pi/4$

(C) $\pi/2$

(D) $2\pi/3$

Q.3

If $\sin^{-1}(x-1) + \cos^{-1}(x-3) + \tan^{-1}\left(\frac{x}{2-x^2}\right) = \cos^{-1} k + \pi$, then the value of $k =$

(A) 1

(B) $-1/\sqrt{2}$

(C) $1/\sqrt{2}$

(D) None of these

Q.4

If $\sum_{k=1}^{100} \tan^{-1} \frac{1}{2k^2} = \tan^{-1} \frac{p}{q}$, where p and q are coprime number, then $2p - q$ is -

(A) 101

(B) 99

(C) 201

(D) 98

Q.5

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Choose the correct options -

(A) No. of solution of the equation $\sin^{-1} \mathrm{X} - \cos^{-1}(-\mathrm{x}) = \frac{\pi}{2}$ is one.

(B) Solution set of the equation $\sin^{-1}\left(x^2+4x+3\right) + \cos^{-1}\left(x^2+6x+8\right) = \frac{\pi}{2}$ is $\left\{-\frac{5}{2}\right\}$.

(C) $\sin^{-1}\left(\cos\left(\sin^{-1}\mathrm{X}\right)\right) + \cos^{-1}\left(\sin\left(\cos^{-1}\mathrm{x}\right)\right)$ is equal to π

(D) $2\left[\tan^{-1}1 + \tan^{-1}2 + \tan^{-1}3\right]$ is equal to π.

Q.6

Value of x satisfying the equation $\sin^{-1}\left(2x^2-1\right) + \cos^{-1}\left(2x\sqrt{1-x^2}\right) = 2\cos^{-1}x$, where $x \in [0,1]$, is

(A) 1

(B) $\frac{1}{\sqrt{2}}$

(C) $\frac{\sqrt{3}}{2}$

(D) 0

Q.7

If α_1 and α_2 satisfies $\sin^{-1}\frac{2x}{1+x^2} = \tan^{-1}\frac{2x}{1-x^2}$ and $|\alpha_1 - \alpha_2| < k$, for all α_1 and α_2, then k is equal to

(A) 1

(B) $3/2$

(C) 2

(D) none of these

Q.8

$\sin^{-1}(\sin 3) + \sin^{-1}(\sin 4) + \sin^{-1}(\sin 5) =$

(A) -1

(B) -2

(C) 12

(D) $2-\pi$.

Q.9

Indicate the relation which is/are true -

(A) $\tan\left|\tan^{-1}x\right| = |x|$

(B) $\cot\left|\cot^{-1}x\right| = x$

(C) $\tan^{-1}|\tan x| = |x|$

(D) $\sin\left|\sin^{-1}x\right| = |x|$

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Q.10

Which of the following is a rational number -

(A) $\sin\left(\tan^{-1} 3 + \tan^{-1} \frac{1}{3}\right)$

(B) $\cos\left(\frac{\pi}{2} - \sin^{-1} \frac{3}{4}\right)$

(C) $\log_2\left(\sin\left(\frac{1}{4}\sin^{-1} \frac{\sqrt{63}}{8}\right)\right)$

(D) $\tan\left(\frac{1}{2}\cos^{-1} \frac{\sqrt{5}}{3}\right)$

Q.11

The value of a for which $ax + \sec^{-1}\sqrt{2x^2 - x^4} + \text{cosec}^{-1}\sqrt{2x^2 - x^4} = 0$

(A) $\frac{\pi}{2}$

(B) $-\frac{\pi}{2}$

(C) $\frac{2}{\pi}$

(D) $-\frac{2}{\pi}$

Q.12

Passage

$f(x) = \sin\left\{\cot^{-1}(x+1)\right\} - \cos\left(\tan^{-1} x\right)$

$a = \cos\tan^{-1}\sin\cot^{-1} x$

$b = \cos\left(2\cos^{-1} x + \sin^{-1} x\right)$

Question:

The value of x for which $f(x) = 0$ is -

(A) $-1/2$

(B) 0

(C) $1/2$

(D) 1

Q.13

Passage

$f(x) = \sin\left\{\cot^{-1}(x+1)\right\} - \cos\left(\tan^{-1} x\right)$

$a = \cos\tan^{-1}\sin\cot^{-1} x$

$b = \cos\left(2\cos^{-1} x + \sin^{-1} x\right)$

Question:

If $f(x) = 0$ then a^2 is equal to -

(A) $1/2$

(B) $2/3$

(C) $5/9$

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(D) $9/5$

Q.14

Passage

$f(x) = \sin\left\{\cot^{-1}(x+1)\right\} - \cos\left(\tan^{-1} x\right)$

$a = \cos\tan^{-1}\sin\cot^{-1} x$

$b = \cos\left(2\cos^{-1} x + \sin^{-1} x\right)$

Question:

If $a^2 = 26/51$, then b^2 is equal to -

(A) $1/25$

(B) $24/25$

(C) $25/26$

(D) $50/51$

Q.15

Passage

It is given that $A = \left(\tan^{-1} x\right)^3 + \left(\cot^{-1} x\right)^3$ where $x > 0$ and $B = \left(\cos^{-1} t\right)^2 + \left(\sin^{-1} t\right)^2$ where $t \in \left[0, \frac{1}{\sqrt{2}}\right]$, and $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$ for $-1 \le x \le 1$ and $\tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}$ for all $x \in R$.

Question:

The interval in which A lies is -

(A) $\left[\frac{\pi^3}{7}, \frac{\pi^3}{2}\right)$

(B) $\left[\frac{\pi^3}{32}, \frac{\pi^3}{8}\right)$

(C) $\left(\frac{\pi^3}{40}, \frac{\pi^3}{10}\right)$

(D) None of these

Q.16

Passage

It is given that $A = \left(\tan^{-1} x\right)^3 + \left(\cot^{-1} x\right)^3$ where $x > 0$ and $B = \left(\cos^{-1} t\right)^2 + \left(\sin^{-1} t\right)^2$ where $t \in \left[0, \frac{1}{\sqrt{2}}\right]$, and $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$ for $-1 \le x \le 1$ and $\tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}$ for all $x \in R$

Question:

The maximum value of B is -

(A) $\frac{\pi^2}{8}$

(B) $\frac{\pi^2}{16}$

(C) $\frac{\pi^2}{4}$

(D) None of these

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Q.17

Passage

It is given that $A = \left(\tan^{-1} x\right)^3 + \left(\cot^{-1} x\right)^3$ where $x > 0$ and $B = \left(\cos^{-1} t\right)^2 + \left(\sin^{-1} t\right)^2$ where $t \in \left[0, \frac{1}{\sqrt{2}}\right]$, and $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$ for $-1 \le x \le 1$ and $\tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}$ for all $x \in R$

Question:

If least value of A is λ and maximum value of B is μ then $\cot^{-1} \cot\left(\frac{\lambda - \mu\pi}{\mu}\right) =$

(A) $\frac{\pi}{8}$

(B) $-\frac{\pi}{8}$

(C) $\frac{7\pi}{8}$

(D) $-\frac{7\pi}{8}$

Q.18

Match the column -

Column I	**Column II**
(A) If $\cos^{-1} p + \cos^{-1} q + \cos^{-1} r = \pi$ then $p^2 + q^2 + r^2 + 2pqr =$	(p) 1
(B) If $\tan^{-1}\frac{1}{m} + \tan^{-1}\frac{1}{n} = \tan^{-1}\frac{1}{\lambda}$, where m, n, $\lambda \in N$ then possible values of n ($\lambda = 3$)	(q) 2 (r) 3
(C) Values which are greater than x for which $\sin(\cot^{-1}(1 + x)) = \cos(\tan^{-1} x)$	(s) 4
(D) Values which are equal to or greater than x for which $5\tan^{-1} x + 3\cot^{-1} x = 2\pi$	(t) 5

(A) A-p ; B-q,t ; C-s,t,q ; D-p,q,r,t

(B) A-p ; B-s,t ; C-p,q,r,s,t ; D-p,q,r,s,t

(C) A-r ; B-s,t ; C-p,q,r,s,t ; D-p,q,r,s

(D) A-t ; B-s,q ; C- p,q,r ; D-p,q,r,t

Q.19

The number of solutions of the equation, $\tan^{-1}(4\{x\}) + \cot^{-1}(x + [x]) = \frac{\pi}{2}$, is
(where [ ] denotes greatest integer function and {.} denotes fractional part function)

Q.20

The value of x satisfying the equation $\tan^{-1}\left(2\tan^{-1}\left(\text{cosec}\left(\tan^{-1} x\right) - \cot\left(\tan^{-1} x\right)\right)\right) + \cot^{-1}\left(2\left(\tan^{-1}\frac{1}{3}\right) + \tan^{-1}\left(\frac{1}{7}\right)\right) = \frac{\pi}{2}$, is

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Answers & Solutions

Q.1 Answer:

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Solution:

We have, $\left(\sin^{-1} x\right)^2 + \left(\cos^{-1} x\right)^2 = \frac{5\pi^2}{8}$

$\Rightarrow \left(\sin^{-1} x\right)^2 + \left(\frac{\pi}{2} - \sin^{-1} x\right)^2 = \frac{5\pi^2}{8} \Rightarrow 2\left(\sin^{-1} x\right)^2 - \pi \sin^{-1} x - \frac{3\pi^2}{8} = 0$

$\Rightarrow \sin^{-1} x = \frac{\pi \pm \sqrt{4\pi^2}}{4} \Rightarrow \sin^{-1} x = \frac{3\pi}{4}$ **or** $-\frac{\pi}{4}$

$\Rightarrow \quad x = \sin\left(-\frac{\pi}{4}\right) \therefore x = -\frac{1}{\sqrt{2}}$

Q.2 Answer:

$\pi/4$

Solution:

$\mathrm{T_n} = \tan^{-1}\left(\frac{1}{\mathrm{n^2+n+1}}\right) = \tan^{-1}\left(\frac{(\mathrm{n+1})-\mathrm{n}}{1+(\mathrm{n+1})\mathrm{n}}\right) = \tan^{-1}(\mathrm{n}+1) - \tan^{-1}\mathrm{n}$

$\Rightarrow \mathrm{T_1} = \tan^{-1}(1/3) = \tan^{-1} 2 - \tan^{-1} 1$

$\mathrm{T_2} = \tan^{-1}(1/7) = \tan^{-1} 3 - \tan^{-1} 2$

$\mathrm{T_n} = \tan^{-1}(\mathrm{n}+1) - \tan^{-1}\mathrm{n}$

On adding,

$\mathrm{T_1} + \mathrm{T_2} + \mathrm{T_3} + \ldots.. + \mathrm{T_n} = \tan^{-1}(\mathrm{n}+1) - \tan^{-1} 1 = \tan^{-1}\left(\frac{\mathrm{n}}{\mathrm{n+2}}\right)$

$\therefore \quad \lim_{n\to\infty}\left(T_1 + T_2 + T_3 + \ldots.. + T_n\right) = \lim_{n\to\infty} \tan^{-1}\left(\frac{1}{1+2/n}\right) = \tan^{-1} 1 = \frac{\pi}{4}$

Q.3 Answer:

$1/\sqrt{2}$

Solution:

$\sin^{-1}(x-1) \Rightarrow -1 \le x-1 \le 1 \Rightarrow 0 \le x \le 2$

$\cos^{-1}(x-3) \Rightarrow -1 \le x-3 \le 1 \Rightarrow 2 \le x \le 4$

$\tan^{-1}\left(\frac{\mathrm{x}}{2-\mathrm{x^2}}\right) \Rightarrow \mathrm{x} \in \mathrm{R}, \quad \mathrm{x} \neq \sqrt{2}, -\sqrt{2}$

$\sin^{-1}(2-1) + \cos^{-1}(2-3) + \tan^{-1}\frac{2}{2-4} = \cos^{-1}\mathrm{k} + \pi$

$\Rightarrow \sin^{-1} 1 + \cos^{-1}(-1) + \tan^{-1}(-1) = \cos^{-1}\mathrm{k} + \pi$

$\frac{\pi}{2} + \pi - \frac{\pi}{4} = \cos^{-1}\mathrm{k} + \pi$

$\Rightarrow \cos^{-1}\mathrm{k} = \frac{\pi}{4} \Rightarrow \mathrm{k} = \frac{1}{\sqrt{2}}$

Q.4 Answer:

99

Solution:

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$\sum \tan^{-1}\left(\frac{2}{1+4k^2-1}\right) = \sum \tan^{-1}\left(\frac{2}{1+(2k-1)(2k+1)}\right)$

$= \sum \tan^{-1}\left(\frac{(2k+1)-(2k-1)}{1+(2k-1)(2k+1)}\right) = \sum_{k=1}^{100} \tan^{-1}(2k+1) - \tan^{-1}(2k-1)$

$= \left(\tan^{-1}(3) - \tan^{-1}(1)\right) + \left(\tan^{-1}(5) - \tan^{-1}(3)\right) + \ldots \left(\tan^{-1}(201) - \tan^{-1}(199)\right)$

$= \tan^{-1}(201) - \tan^{-1}(1) = \tan^{-1}\left(\frac{201-1}{1+201}\right) = \tan^{-1}\left(\frac{200}{202}\right) = \tan^{-1}\left(\frac{100}{101}\right)$

$2p - q = 200 - 101 = 99$

Q.5 Answer:

Solution set of the equation $\sin^{-1}\left(x^2+4x+3\right) + \cos^{-1}\left(x^2+6x+8\right) = \frac{\pi}{2}$ is $\left\{-\frac{5}{2}\right\}$.

Solution:

(A) $\sin^{-1} x - \frac{\pi}{2} + \sin^{-1}(-x) = \frac{\pi}{2} \Rightarrow \sin^{-1}(x) + \sin^{-1}(-x) = \pi$

$0 = \pi$ **which is not possible.**

$\therefore$ no solution.

(B) $\sin^{-1}\left(x^2+4x+3\right) + \cos^{-1}\left(x^2+6x+8\right) = \frac{\pi}{2}$

$= \sin^{-1}\left(x^2+4x+3\right) + \cos^{-1}\left(x^2+4x+3\right)$

$\Rightarrow x^2+6x+8 = x^2+4x+3 \Rightarrow 2x = -5 \Rightarrow x = -\frac{5}{2}$

$x^2+4x+3 = (x+2)^2 - 1 \in [-1,1]$ at $x = -\frac{5}{2}$

and $x^2+6x+8 = (x+3)^2 - 1 \in [-1,1]$ **at** $x = -\frac{5}{2}$

$\therefore \quad \mathrm{x} = -\frac{5}{2}$

(C) $-1 \le \cos\left(\sin^{-1}\mathrm{X}\right) \le 1$ **and** $-1 \le \sin\left(\cos^{-1}\mathrm{X}\right) \le 1$,

$\sin^{-1}\left\{\cos\left(\sin^{-1}\mathrm{x}\right)\right\} + \cos^{-1}\left(\sin\left(\cos^{-1}\mathrm{x}\right)\right\} = \frac{\pi}{2}$

(D) $2\left[\tan^{-1}\frac{1+2}{1-2} + \pi + \tan^{-1}3\right] = 2\left[\pi - \tan^{-1}3 + \tan^{-1}3\right] = 2\pi$

Q.6 Answer:

$\frac{\sqrt{3}}{2}$

Solution:

$x \in [0,1]$

$\sin^{-1}\left(2x^2-1\right) + \cos^{-1}\left(2x\sqrt{1-x^2}\right) = 2\cos^{-1}x$

Let $\mathrm{x} = \cos\theta, \theta \in \left[0, \frac{\pi}{2}\right]$

$\sin^{-1}(\cos 2\theta) + \cos^{-1}(\sin 2\theta) = 2\cos^{-1}(\cos\theta)$

If $\theta \in \left[0, \frac{\pi}{4}\right]$

$\frac{\pi}{2} - 2\theta + \frac{\pi}{2} - 2\theta = 2\theta$

$\Rightarrow \quad 6\theta = \pi$

$\Rightarrow \quad \theta = \frac{\pi}{6}$

$\cos\theta = \frac{\sqrt{3}}{2}$

$\Rightarrow \quad \mathrm{x} = \frac{\sqrt{3}}{2}$

If $\theta \in \left[\frac{\pi}{4}, \frac{\pi}{2}\right]$

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$\frac{\pi}{2} - 2\theta + 2\theta - \frac{\pi}{2} = 2\theta$

$\Rightarrow \quad \theta = 0$

Not possible

Q.7 Answer:

2

Solution:

graph of $\sin^{-1}\left(\frac{2x}{1+x^2}\right)$

graph of $\tan^{-1}\left(\frac{2x}{1-x^2}\right)$

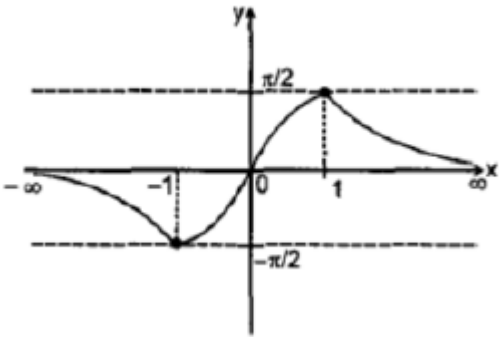
from graph one can say

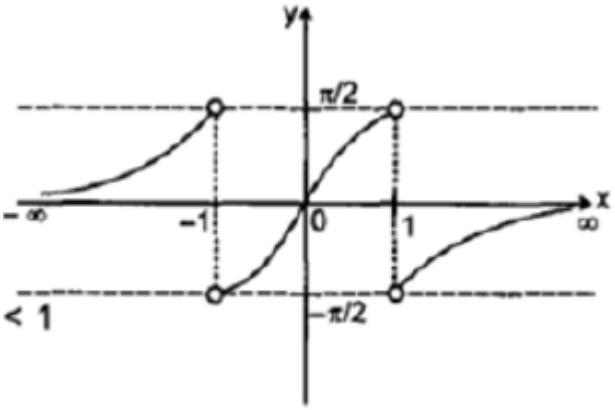
$-1 < x < 1 \Rightarrow -1 < \alpha_1 < 1$

and $\quad -1 < \alpha_2 < 1 \Rightarrow |\alpha_1 - \alpha_2| < 2$

$-1 < x < 1 \Rightarrow -1 < \alpha_1 < 1 \quad$ and $\quad -1 < \alpha_2 < 1$

$\Rightarrow \quad |\alpha_1 - \alpha_2| < 2$





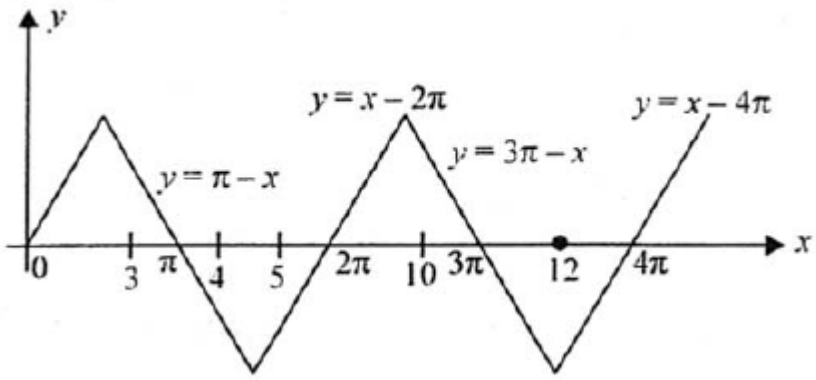
Q.8 Answer:

-2

Solution:

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The graph of $y = \sin^{-1}(\sin x)$ is



$y(3) + y(4) + y(5) = (\pi - 3) + (\pi - 4) + (5 - 2\pi) = -2$

Q.9 Answer:

$\tan\left|\tan^{-1} x\right| = |x|$

Solution:

Since $\left|\tan^{-1} x\right| = \begin{cases} \tan^{-1} x & \text{if } 0 \le \tan^{-1} x < \pi/2 \\ -\tan^{-1} x & \text{if } -\pi/2 < \tan^{-1} x < 0 \end{cases}$

$\begin{cases} \tan^{-1} x & \text{if } x \ge 0 \\ -\tan^{-1} x & \text{if } x < 0 \end{cases} \Rightarrow \left|\tan^{-1} x\right| = \tan^{-1}|x| \;\forall\, x \in R$

$\Rightarrow \quad \tan\left|\tan^{-1} x\right| = \tan\tan^{-1}|x| = |x|$

Likewise $\sin\left|\sin^{-1} x\right| = \sin\sin^{-1}|x| = |x| \quad \forall |x| \le 1$

$\cot\left|\cot^{-1} x\right|$ as $0 < \left|\cot^{-1} x\right| < \pi \quad \forall x \in R$

$\Rightarrow \quad \cot\left|\cot^{-1} x\right| = \cot\cot^{-1} x = x$

$|\tan x|$ is not necessarily always equal to $\tan|x|$

Q.10 Answer:

$\sin\left(\tan^{-1} 3 + \tan^{-1}\frac{1}{3}\right)$

Solution:

(A) $\sin\left(\tan^{-1} 3 + \tan^{-1}\frac{1}{3}\right) = \sin\frac{\pi}{2} = 1$

(B) $\cos\left(\frac{\pi}{2} - \sin^{-1}\frac{3}{4}\right) = \cos\left(\cos^{-1}\frac{3}{4}\right) = \frac{3}{4}$

(C) $\sin\left(\frac{1}{4}\sin^{-1}\frac{\sqrt{63}}{8}\right)$

Let $\sin^{-1}\frac{\sqrt{63}}{8} = \theta$

So, $\sin\theta = \frac{\sqrt{63}}{8}$ **if** $\cos\theta = 1/8$

We have $\cos\frac{\theta}{2} = \sqrt{\frac{1+\cos\theta}{2}} = \frac{3}{4}; \sin\frac{\theta}{4} = \sqrt{\frac{1-\cos\theta/2}{2}} = \frac{1}{2\sqrt{2}}$

Now, $\log_2\left(\sin\left(\frac{1}{4}\sin^{-1}\frac{\sqrt{63}}{8}\right)\right) = \log_2\frac{1}{2\sqrt{2}} = -\frac{3}{2}$

(D) $\cos^{-1}\frac{\sqrt{5}}{3} = \theta; \cos\theta = \frac{\sqrt{5}}{3}$

$\therefore \quad \tan\frac{\theta}{2} = \frac{3-\sqrt{5}}{2}$ **which is irrational.**

Q.11 Answer:

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$\frac{\pi}{2}$

Solution:

For existence of $\sec^{-1}\left(\sqrt{2x^2-x^4}\right)$ **or** $\operatorname{cosec}^{-1}\left(\sqrt{2x^2-x^4}\right)$

$2x^2-x^4 \geq 1$ **i.e.** $\left(x^2-1\right)^2 \leq 0$

hence $x^2=1 \Rightarrow x=\pm 1$

if $x=1, a+\frac{\pi}{2}=0$

if $x=-1, -a+\frac{\pi}{2}=0 \Rightarrow a=\pm\frac{\pi}{2}$

Q.12 Answer:

$-1/2$

Solution:

$\mathrm{f(x)}=0 \Rightarrow \sin\left\{\cot^{-1}(\mathrm{x}+1)\right\}-\cos\left(\tan^{-1}\mathrm{x}\right)$

$\Rightarrow \sin\sin^{-1}\frac{1}{\sqrt{1+(x+1)^2}}=\operatorname{coscos}^{-1}\frac{1}{\sqrt{1+x^2}}$

$\Rightarrow \frac{1}{\sqrt{1+(x+1)^2}}=\frac{1}{\sqrt{1+x^2}} \Rightarrow 1+x^2=2+x^2+2x$

$x=-1/2$, **so** $f(x)=0$ **for** $x=-1/2$

$\mathrm{a}=\cos\tan^{-1}\sin\cot^{-1}\mathrm{x}=\cos\tan^{-1}\sin\alpha$

where $\mathrm{x}=\cot\alpha$

$=\cos\tan^{-1}\frac{1}{\sqrt{1+x^2}}=\cos\beta$ **where** $\tan\beta=\frac{1}{\sqrt{1+x^2}}$

$=\sqrt{\frac{x^2+1}{x^2+2}} \Rightarrow a^2=\frac{x^2+1}{x^2+2}=\frac{5}{9}$ **for** $x=-1/2$

Now $a^2=26/51 \Rightarrow x=\pm 1/5$

and $b=\cos\left(2\cos^{-1}x+\sin^{-1}x\right)=\cos\left(\cos^{-1}x+\pi/2\right)$

$\mathrm{b}=-\sin\left(\cos^{-1}\mathrm{X}\right)=-\sqrt{1-\mathrm{x}^2}$

$b^2=1-x^2=24/25$ **for** $x=\pm(1/5)$

Q.13 Answer:

$5/9$

Solution:

$\mathrm{f(x)}=0 \Rightarrow \sin\left\{\cot^{-1}(\mathrm{x}+1)\right\}-\cos\left(\tan^{-1}\mathrm{x}\right)$

$\Rightarrow \quad \sin\sin^{-1}\frac{1}{\sqrt{1+(\mathrm{x}+1)^2}}=\cos\cos^{-1}\frac{1}{\sqrt{1+\mathrm{x}^2}}$

$\Rightarrow \quad \frac{1}{\sqrt{1+(\mathrm{x}+1)^2}}=\frac{1}{\sqrt{1+\mathrm{x}^2}} \Rightarrow 1+\mathrm{x}^2=2+\mathrm{x}^2+2\mathrm{x}$

$\mathrm{x}=-1/2$, **so** $\mathrm{f(x)}=0$ **for** $\mathrm{x}=-1/2$

$\mathrm{a}=\cos\tan^{-1}\sin\cot^{-1}\mathrm{x}=\cos\tan^{-1}\sin\alpha$ **where** $\mathrm{x}=\cot\alpha$

$=\cos\tan^{-1}\frac{1}{\sqrt{1+\mathrm{x}^2}}=\cos\beta$ **where** $\tan\beta=\frac{1}{\sqrt{1+\mathrm{x}^2}}$

$=\sqrt{\frac{\mathrm{x}^2+1}{\mathrm{x}^2+2}} \Rightarrow \mathrm{a}^2=\frac{\mathrm{x}^2+1}{\mathrm{x}^2+2}=\frac{5}{9}$ **for** $\mathrm{x}=-1/2$

Now $\mathrm{a}^2=26/51 \Rightarrow \mathrm{x}=\pm 1/5$

and $\mathrm{b}=\cos\left(2\cos^{-1}\mathrm{x}+\sin^{-1}\mathrm{x}\right)=\cos\left(\cos^{-1}\mathrm{x}+\pi/2\right)$

$\mathrm{b}=-\sin\left(\cos^{-1}\mathrm{x}\right)=-\sqrt{1-\mathrm{x}^2}$

$\mathrm{b}^2=1-\mathrm{x}^2=24/25$ **for** $\mathrm{x}=\pm(1/5)$

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Q.14 Answer:

$24/25$

Solution:

$f(x) = 0 \Rightarrow \sin\left\{\cot^{-1}(x+1)\right\} - \cos\left(\tan^{-1} x\right)$

$\Rightarrow \quad \sin\sin^{-1}\frac{1}{\sqrt{1+(x+1)^2}} = \cos\cos^{-1}\frac{1}{\sqrt{1+x^2}}$

$\Rightarrow \quad \frac{1}{\sqrt{1+(x+1)^2}} = \frac{1}{\sqrt{1+x^2}} \Rightarrow 1 + x^2 = 2 + x^2 + 2x$

$x = -1/2$, **so** $f(x) = 0$ **for** $x = -1/2$

$a = \cos\tan^{-1}\sin\cot^{-1} x = \cos\tan^{-1}\sin\alpha$

where $x = \cot\alpha$

$= \cos\tan^{-1}\frac{1}{\sqrt{1+x^2}} = \cos\beta$

where $\tan\beta = \frac{1}{\sqrt{1+x^2}}$

$= \sqrt{\frac{x^2+1}{x^2+2}} \Rightarrow a^2 = \frac{x^2+1}{x^2+2} = \frac{5}{9}$ **for** $x = -1/2$

Now $a^2 = 26/51 \Rightarrow x = \pm 1/5$

and $b = \cos\left(2\cos^{-1} x + \sin^{-1} x\right) = \cos\left(\cos^{-1} x + \pi/2\right)$

$b = -\sin\left(\cos^{-1} x\right) = -\sqrt{1 - x^2}$

$b^2 = 1 - x^2 = 24/25$ **for** $x = \pm(1/5)$

Q.15 Answer:

$\left[\frac{\pi^3}{32}, \frac{\pi^3}{8}\right)$

Solution:

$A = \left(\tan^{-1} x\right)^3 + \left(\cot^{-1} x\right)^3$

$A = \left(\tan^{-1} x + \cot^{-1} x\right)^3 - 3\tan^{-1} x\cot^{-1} x\left(\tan^{-1} x + \cot^{-1} x\right)$

$\Rightarrow \quad A = \left(\frac{\pi}{2}\right)^3 - 3\tan^{-1} x\cot^{-1} x\frac{\pi}{2} = \frac{\pi^3}{8} - \frac{3\pi}{2}\tan^{-1} x\left(\frac{\pi}{2} - \tan^{-1} x\right)$

$\Rightarrow \quad A = \frac{\pi^3}{32} + \frac{3\pi}{2}\left(\tan^{-1} x - \frac{\pi}{2}\right)^2$ **as** $x > 0$

$\frac{\pi^3}{32} \le A < \frac{\pi^3}{8}$

Q.16 Answer:

$\frac{\pi^2}{4}$

Solution:

$B = \left(\cos^{-1} t\right)^2 + \left(\sin^{-1} t\right)^2$

$B = \left(\sin^{-1} t + \cos^{-1} t\right)^2 - 2\sin^{-1} t\cos^{-1} t$

$B = \frac{\pi^2}{4} - 2\sin^{-1} t\left(\frac{\pi}{2} - \sin^{-1} t\right) = \frac{\pi^2}{8} + 2\left(\sin^{-1} t - \frac{\pi}{4}\right)^2; B_{max} = \frac{\pi^2}{8} + 2\frac{\pi^2}{16} = \frac{\pi^2}{4}$

Q.17 Answer:

$\frac{\pi}{8}$

Solution:

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$\lambda = \frac{\pi^3}{32}, \mu = \frac{\pi^2}{4}; \frac{\lambda}{\mu} = \frac{\pi}{8}; \frac{\lambda - \mu\pi}{\mu} = \frac{\pi}{8} - \pi = \frac{-7\pi}{8}$

$\cot^{-1}\cot\left(\frac{\lambda - \mu\pi}{\mu}\right) = \cot^{-1}\cot\left(-\frac{7\pi}{8}\right) = \frac{\pi}{8}$

Q.18 Answer:

A-p ; B-s,t ; C-p,q,r,s,t ; D-p,q,r,s,t

Solution:

(A) - p, (B) - st, (C) - pqrst, (D) - pqrst

(A) According to given condition, we put $p = q = r = \frac{1}{2}$

Then, $p^2 + q^2 + r^2 + 2pqr = \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2 + 2 \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4} + \frac{1}{4} + \frac{1}{4} + \frac{2}{8} = 1$

(B) $\tan^{-1}\frac{1}{m} + \tan^{-1}\frac{1}{n} = \tan^{-1}\frac{1}{\lambda}$

$\tan^{-1}\left(\frac{\frac{1}{m}+\frac{1}{n}}{1-\frac{1}{m}\cdot\frac{1}{n}}\right) = \tan^{-1}\frac{1}{\lambda} \Rightarrow m = \frac{n\lambda+1}{n-\lambda} = \lambda + \frac{1+\lambda^2}{n-\lambda}$

$\lambda = 3; m = 3 + \dfrac{10}{n-3} \Rightarrow n = 4, 5, 8, 13$

Hence, $(m, n) \Rightarrow (4, 13), (13, 4), (8, 5), (5, 8)$

(C) $\sin\left(\cot^{-1}(1+x)\right) = \cos\left(\tan^{-1}X\right)$

If $\alpha = \cot^{-1}(1+x)$ **and** $\beta = \tan^{-1}X$

Then, $\frac{1}{\sqrt{x^2+2x+2}} = \frac{1}{\sqrt{1+x^2}} \Rightarrow x = -1/2$

(D) $3\left(\tan^{-1}x + \cot^{-1}x\right) + 2\tan^{-1}x = 2\pi$

$\tan^{-1}x = \frac{\pi}{4} \Rightarrow x = 1$ **is solution**

Q.19 Answer:

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Solution:

$4\{x\} = x + [x]$

$= x + x - \{x\} \Rightarrow 5\{x\} = 2x \Rightarrow \{x\} = \frac{2x}{5}$

$\Rightarrow \quad 0 \le \frac{2x}{5} < 1 \Rightarrow 0 \le x < \frac{5}{2},$ hence, $[x] = 0, 1, 2$

Again from (1), $4x - 4[x] = x + [x]$

$3x = 5[x]$

Case-I : If $x \in [0, 1) \Rightarrow [x] = 0$

$\therefore \quad 3x = 0 \Rightarrow x = 0$

Case-II : If $x \in [1, 2) \Rightarrow [x] = 1$

$\therefore \quad 3x = 5 \Rightarrow x = 5/3$

Case-III : If $x \in (2, 5/2] \in [x] = 2$

$\therefore \quad 3x = 10 \Rightarrow x = 10/3$ **(reject)**

$\therefore \quad$ **Number of solutions** $= 2$

Q.20 Answer:

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Solution:

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$2\tan^{-1}\left(\operatorname{cosec}\left(\tan^{-1}x\right)-\cot\left(\tan^{-1}x\right)\right)$

$=2\tan^{-1}\frac{1}{3}+\tan^{-1}\frac{1}{7}$

$\text{LHS }=2\tan^{-1}\left(\operatorname{cosec}\left(\sin^{-1}\left(\frac{x}{\sqrt{1+x^2}}\right)-\cot^{-1}\left(\cot^{-1}\frac{1}{x}\right)\right)\right)$

$=2\tan^{-1}\left(\frac{\sqrt{1+x^2}}{x}-\frac{1}{x}\right)$

$\Rightarrow\tan^{-1}\left(\frac{2\left(\sqrt{1+x^2}-1\right)x}{x^2-\left(\sqrt{1+x^2}-1\right)^2}\right)=\tan^{-1}x$

$\text{RHS }=\tan^{-1}\left(\frac{3}{4}\right)+\tan^{-1}\frac{1}{7}=\tan^{-1}1$

$\Rightarrow\tan^{-1}x$

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